

MATHEMATICS 2

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **26 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 2

- 1 Find the sum of all solutions θ in the interval $0 \leq \theta < 2\pi$ to the equation:

$$\frac{2 \sin^2 \theta + \sin \theta - 1}{\sin \theta + 1} = 0$$

- A $\frac{\pi}{6}$
- B π
- C $\frac{3\pi}{2}$
- D $\frac{5\pi}{2}$
- E 3π
- F 0

- 2 The diagram shows a trapezium $OABC$ in the first quadrant ($x \geq 0, y \geq 0$).

The vertex O is at the origin $(0, 0)$, vertex A has coordinates $(2a, a)$, vertex B has coordinates $(a, 3a)$, and vertex C lies on the positive y -axis, where $a > 0$.

The side CB is parallel to the side OA .

The area of the trapezium $OABC$ is 60 square units.

What are the coordinates of vertex B ?

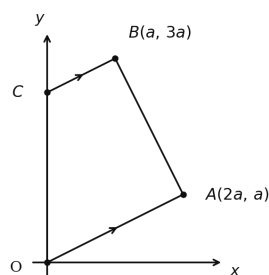


Diagram not to scale

- A $(2, 6)$
- B $(2\sqrt{2}, 6\sqrt{2})$
- C $(2\sqrt{5}, 6\sqrt{5})$
- D $(4, 12)$
- E $(6, 18)$
- F $(8, 4)$

- 3 An arithmetic progression has first term 100 and common difference -6.

What is the maximum possible value of the sum of the first n terms?

- A 4
- B 850
- C 880
- D 882
- E 884
- F 901

- 4 A curve passes through the origin and for $x > 0$ its gradient is given by

$$\frac{dy}{dx} = 3x^2$$

At a point P on the curve, the gradient is m , where m is a positive constant.

What is the y-coordinate of P in terms of m ?

- A $\left(\frac{m}{3}\right)^{\frac{3}{2}}$
- B $\frac{m^3}{27}$
- C $\left(\frac{3}{m}\right)^{\frac{3}{2}}$
- D $\frac{m^2}{12}$
- E m
- F $\left(\frac{m^3}{3}\right)^{\frac{1}{2}}$

- 5 A monic cubic curve $y = f(x)$ has stationary points at $x = 1$ and $x = 3$.
Which one of the following conditions is **not** sufficient to determine the equation of the curve uniquely?
- A The curve passes through the point $(2, 5)$.
 - B The y -coordinate of the local maximum is 1.
 - C The curve passes through the point $(2, 1)$.
 - D The value of $\int_0^1 f(x) \, dx$ is 4.
 - E The equation $f(x) = 0$ has exactly two distinct real roots.
 - F The tangent to the curve at $x = 0$ passes through the point $(1, 5)$.
- 6 A quantity is modelled by a function $f(x)$ that exhibits exponential decay. For a fixed positive constant d , increasing x by d causes the value of $f(x)$ to decrease by 20%.
What is the total percentage decrease in the value of $f(x)$ when x is increased by $2d$?
- A 36%
 - B 40%
 - C 64%
 - D 4%
 - E 44%
 - F 16%
- 7 Variables x and y are related by the equation $y = Ab^{2x+3}$, where A and b are constants.
A graph of $\ln y$ against x is a straight line with gradient 4 and intercept 7 on the vertical axis.
Which of the following is the value of A ?
- A e^{-5}
 - B e
 - C e^2
 - D e^4
 - E e^7

- 8 What is the sum of the real roots of the equation

$$\left(\frac{x^2}{x+1}\right)^2 - 4\left(\frac{x^2}{x+1}\right) = 5?$$

- A -4
 - B 1
 - C 4
 - D 5
 - E 6
- 9 A circle is tangent to both the x -axis and the y -axis. The circle passes through the point $(1, 2)$. What is the radius of the largest such circle?
- A 1
 - B 2.5
 - C 3
 - D 5
 - E $\sqrt{5}$
- 10 Which of the following equations has no real solutions?
- A $x^3 + x = 3$
 - B $\ln(x) = 2 - x$
 - C $e^x = 2 - x^2$
 - D $\sin(x) = x - 1$
 - E $3^x + 3^{-x} = \frac{3}{2}$

- 11 A function $f(x)$ has derivative $f'(x) = 12x^2 - 12x$.
What is the value of the definite integral

$$\int_0^1 f(x) dx$$

- A -2
- B -1
- C 0
- D 1
- E 2
- F Cannot be determined from the information given

- 12 A geometric sequence T_n is defined for $n = 1, 2, 3, \dots$ by

$$T_n = 8(4^{n-1})$$

A second sequence U_n is defined by $U_n = \log_2(T_n)$.

Find the value of the positive integer N for which the sum of the first N terms of the sequence U_n is 168.

- A 8
- B 11
- C 12
- D 14
- E 16
- F 18

- 13 The set of values of x for which $\frac{6}{x-1} \geq x$ is given by:

- A $1 < x \leq 3$
- B $-2 \leq x \leq 3$
- C $x \leq -2$ or $1 < x \leq 3$
- D $-2 \leq x < 1$ or $x \geq 3$
- E $x \leq -2$ or $x \geq 3$

- 14 A particle undergoes a displacement of 12 units on a bearing of 060° , followed by a displacement of L units on a bearing of 330° .

The final position of the particle is due North of the starting point.

What is the value of L ?

- A $4\sqrt{3}$
- B $6\sqrt{3}$
- C 12
- D $12\sqrt{3}$
- E 24

- 15** The polynomial $f(x) = (x + 3)^{10}$ is expanded in powers of x .
How many of the coefficients in this expansion are divisible by 3?

A 1
B 9
C 10
D 11
E 5

- 16** The diagram shows a trapezium and a right-angled triangle, with dimensions given in terms of x .
All given lengths must be strictly positive.

Find the complete set of values of x for which the area of the trapezium is strictly greater than the area of the triangle.

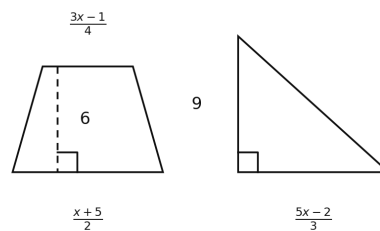


Diagram not to scale

- A** $x < \frac{13}{5}$
B $x > \frac{13}{5}$
C $\frac{1}{3} < x < \frac{13}{5}$
D $\frac{2}{5} < x < \frac{13}{5}$
E $x < \frac{4}{3}$
F $\frac{1}{3} < x < \frac{4}{3}$
G $\frac{2}{5} < x < \frac{4}{3}$
H $\frac{2}{5} < x < \frac{31}{15}$

- 18 A geometric series has first term $\ln(\sqrt{2}e \cos x)$ and common ratio $\sin^2 x$, for an angle x in the range $0 < x < \frac{\pi}{2}$.

The sum to infinity of the series is 1.

What is the value of x ?

- A $\frac{\pi}{6}$
- B $\frac{\pi}{4}$
- C $\frac{\pi}{3}$
- D $\frac{\pi}{2}$
- E $\arccos\left(\frac{1}{\sqrt{2}e}\right)$

- 19 The diagram shows a rectangle of length $\sqrt{5} + \sqrt{2}$ and width $\sqrt{5} - \sqrt{2}$. A square of side length $\sqrt{5} - \sqrt{2}$ is removed from one end, leaving the shaded region.

What is the area of the shaded region?

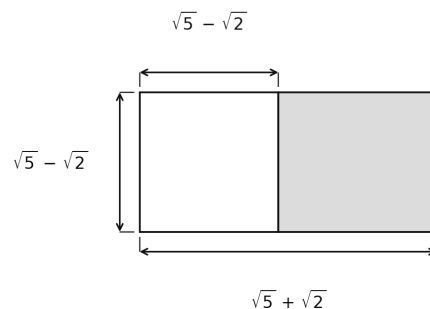


Diagram not to scale

- A $2\sqrt{10} - 4$
- B $2\sqrt{10} + 4$
- C $\sqrt{10} - 4$
- D $4 - 2\sqrt{10}$
- E $7 - 2\sqrt{10}$

- 20** The curve with equation $y = x^3 - 3x^2$ and the line with equation $y = 9x + k$ intersect at exactly three distinct points.

Which of the following describes all possible values of the constant k ?

- A** $-27 < k < 5$
- B** $-27 \leq k \leq 5$
- C** $-5 < k < 27$
- D** $-5 \leq k \leq 27$
- E** $k < -27$ or $k > 5$

- 21** Let x and y be positive real numbers such that the expression below is defined.

$$\frac{2 - \log_x(y)}{(\log_x(y))^2 + \log_x(y) - 6} = \log_k(x)$$

Find k in terms of x and y .

- A** $\frac{1}{yx^3}$
- B** yx^3
- C** $\frac{x^3}{y}$
- D** $\frac{y}{x^3}$

- 22** A function $f(x)$ is defined by

$$f(x) = \frac{2x^3 + 2x^2 - 12x}{x^2 + x - 6}$$

What is the coefficient of x^3 in the expansion of $(1 - f(x))^5$?

- A** 80
- B** -40
- C** -80
- D** -10
- E** 10
- F** -480

- 23** A geometric progression $x_1, x_2, x_3, \dots$ consists of positive real numbers.

Let S be the sum $S = \sum_{n=1}^{15} \log_b(x_n)$, where $b > 1$.

Which one of the following pieces of information is sufficient to determine the value of S ?

- A** The value of $x_7 + x_9$ is known.
- B** The value of $x_4 x_{12}$ is known.
- C** The value of $\sum_{n=1}^{15} x_n$ is known.
- D** The value of $x_1 x_{16}$ is known.
- E** The value of $\prod_{n=1}^{14} x_n$ is known.

- 24** A triangle has vertices at $(0, 0)$, $(x, 0)$, and $(x, f(x))$ for $x > 0$. The area of the triangle, A , is a function of x .

The rate of change of the area with respect to x is given by

$$\frac{dA}{dx} = 2 \sin^2(x)$$

Given that $A = 0$ when $x = 0$, what is the value of $f\left(\frac{\pi}{4}\right)$?

- A** $2 - \frac{4}{\pi}$
- B** $\frac{8}{\pi}$
- C** $1 - \frac{2}{\pi}$
- D** $2 + \frac{4}{\pi}$
- E** $1 + \frac{2}{\pi}$
- F** $\frac{16}{\pi}$

- 25 Which one of the following is a simplification of $5 - \frac{9 - 4x^2}{2x^2 - 3x}$ for all valid values of x ?

- A $3 - \frac{3}{x}$
- B $3 + \frac{3}{x}$
- C $5 - \frac{3}{x}$
- D $5 + \frac{3}{x}$
- E $7 - \frac{3}{x}$
- F $7 + \frac{3}{x}$

- 26 Find the exact value of the sum

$$\sum_{k=1}^{10} \log_3 \left(\frac{\sqrt{3} \times 9^k}{27^{k-1}} \right)$$

- A -95
- B -85
- C -75
- D -70
- E -55
- F 195

- 27 A function f is defined for $x > 0$ by

$$f(x) = \frac{(2x^2)^{-3}}{x^{-10}}$$

What is the value of $f'(1)$?

- A $\frac{1}{2}$
- B -32
- C -2
- D $\frac{1}{8}$
- E 8
- F $\frac{15}{2}$